

1. Launch one aws ec2 and install jenkins.
2. Setup Apache Tomcat on Ubuntu Virtual Machine
3. Setup Docker Desktop on your local windows.

Instances (1/1) [Info](#)

Connect

Instance state ▼

Actions ▼

Launch instance

All states ▼

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	server tomcat	i-04691240af52436b7	Running	t3.small	3/3 checks passed View alarms +		eu-north-1c	ec2-13-50-240-232.eu-...

```
waseemakram@waseem downloads % ssh -i waseem.pem ec2-user@13.50.240.232
```

A newer release of "Amazon Linux" is available.
Version 2023.9.20251020:
Version 2023.9.20251027:
Run "/usr/bin/dnf check-release-update" for full release and version update info

```
,      #_
~\     #####_       Amazon Linux 2023
~~   \_#####\
~~       \###|
~~         \#/ ____ https://aws.amazon.com/linux/amazon-linux-2023
~~           V~' '->
~~~~
    ~~-._.-/_
        _/_/_
          /m/'
```

Last login: Thu Oct 30 11:44:21 2025 from 13.48.4.203
[ec2-user@ip-172-31-7-102 ~]\$ ls

Installed jenkins

By yum install Jenkins

```
[ec2-user@ip-172-31-7-102 ~]$ sudo yum install jenkins -y
Jenkins-stable
Dependencies resolved.
615 kB/s | 32 kB

Package Architecture Version Repository
-----
Installing:
jenkins noarch 2.528.1-1.1 jenkins
Transaction Summary
-----
Install 1 Package

Total download size: 91 M
Installed size: 91 M
Downloading Packages:
jenkins-2.528.1-1.1.noarch.rpm
36 MB/s | 91 MB
-----
Total
35 MB/s | 91 MB
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: jenkins-2.528.1-1.1.noarch
Installing : jenkins-2.528.1-1.1.noarch
Running scriptlet: jenkins-2.528.1-1.1.noarch
Verifying : jenkins-2.528.1-1.1.noarch

WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.9.20251020:
Run the following command to upgrade to 2023.9.20251020:

dnf upgrade --releasever=2023.9.20251020

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.9.20251020.html

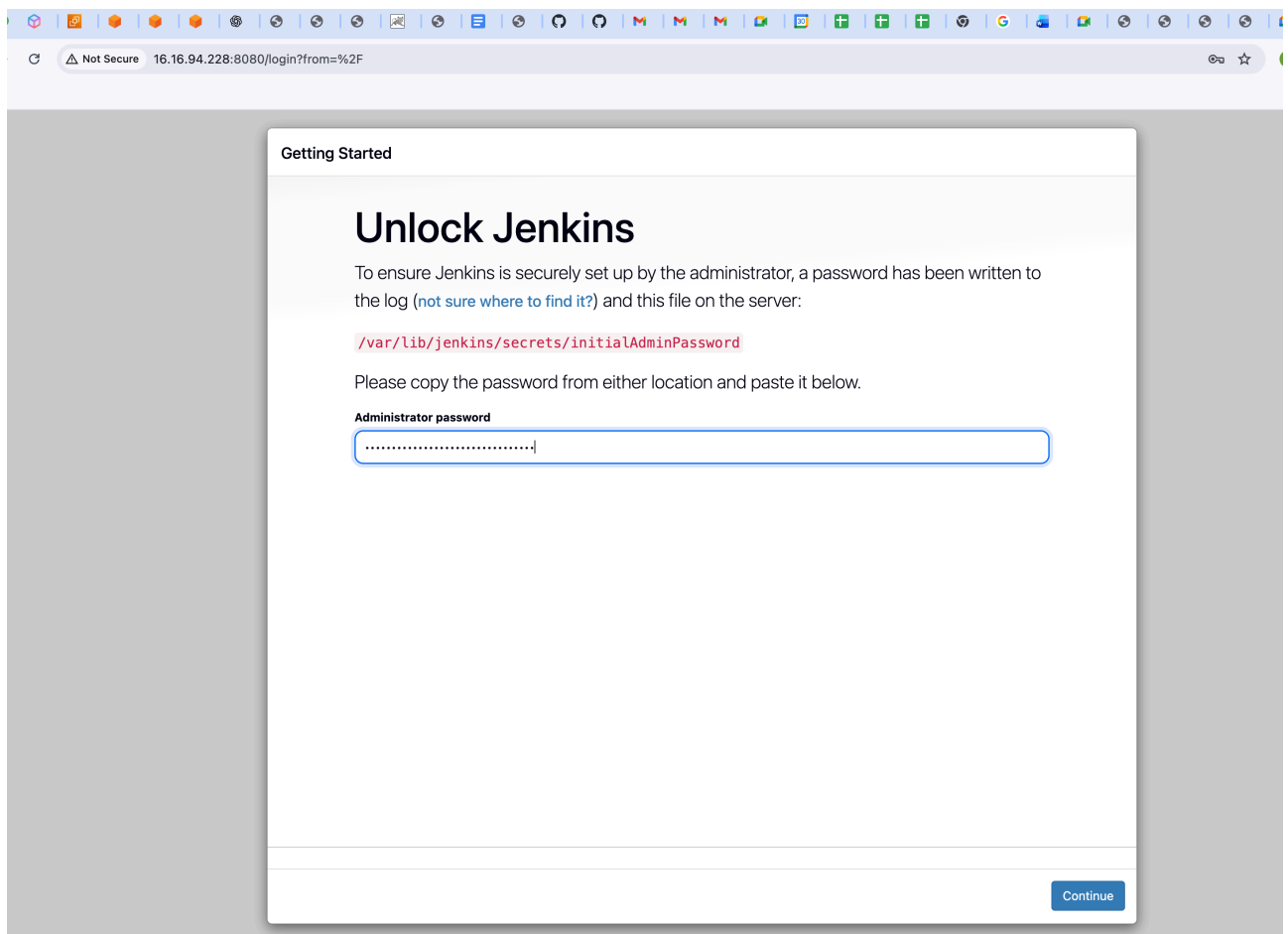
Version 2023.9.20251027:
Run the following command to upgrade to 2023.9.20251027:
```

Sudo systemctl start Jenkins

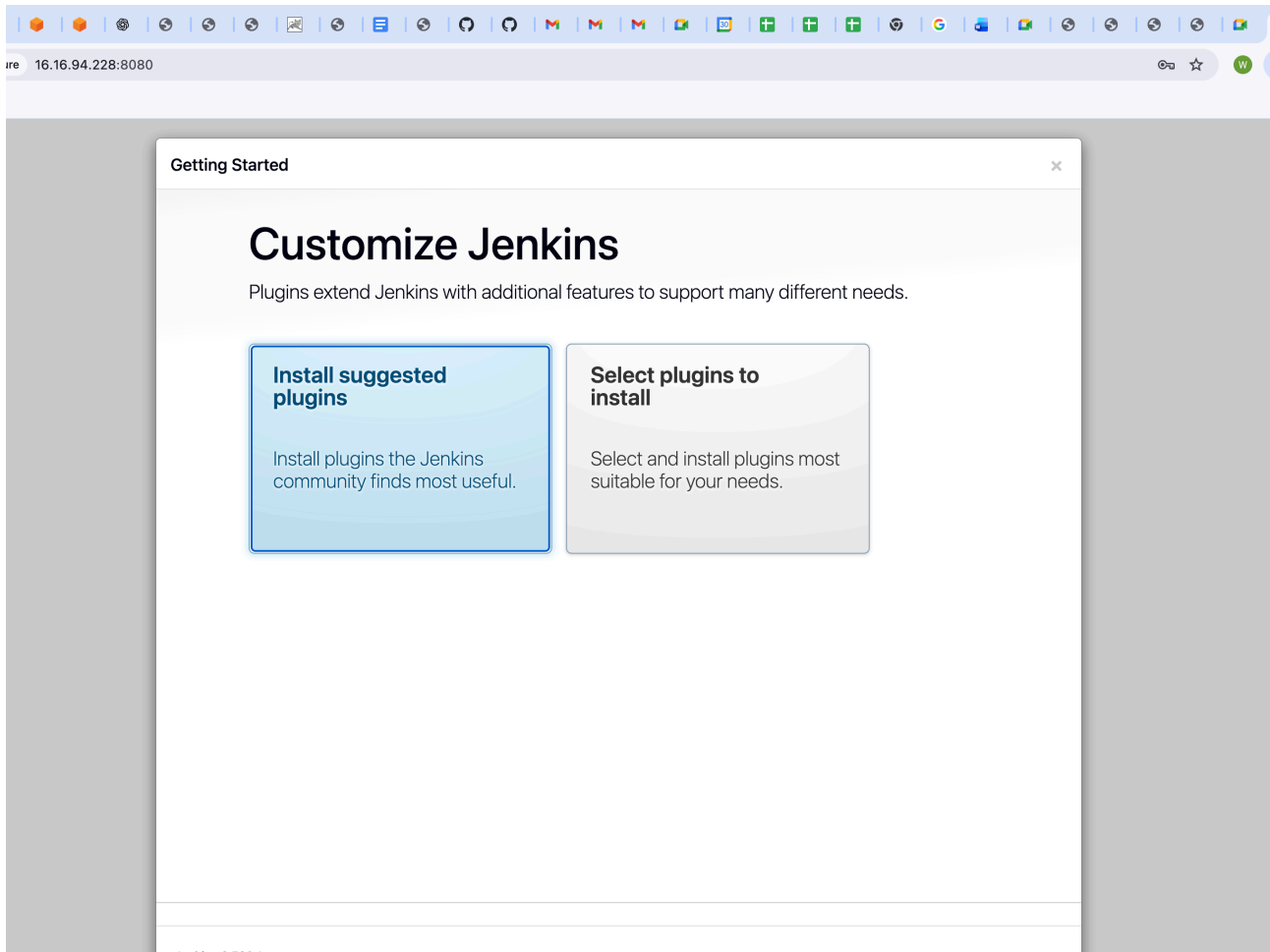
```
Installed:
jenkins-2.528.1-1.1.noarch

Complete!
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl enable jenkins
[sudo systemctl start jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
^[[B^[[A^C
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl enable jenkins
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-10-30 12:59:03 UTC; 43s ago
     Main PID: 2616 (java)
       Tasks: 52 (limit: 2248)
      Memory: 563.8M
         CPU: 23.397s
    CGroup: /system.slice/jenkins.service
            └─2616 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Oct 30 12:58:57 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: [LF]>
Oct 30 12:58:57 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: [LF]> *****
Oct 30 12:58:57 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: [LF]> *****
Oct 30 12:58:57 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: [LF]> *****
Oct 30 12:59:03 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: 2025-10-30 12:59:03.176+0000 [id=30] INFO jenkins.InitReactorRunne
Oct 30 12:59:03 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: 2025-10-30 12:59:03.190+0000 [id=23] INFO hudson.lifecycle.Lifecyc
Oct 30 12:59:03 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 30 12:59:04 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: 2025-10-30 12:59:04.036+0000 [id=49] INFO h.m.DownloadService$Down
Oct 30 12:59:04 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: 2025-10-30 12:59:04.037+0000 [id=49] INFO hudson.util.Retrier#star
Oct 30 12:59:08 ip-172-31-7-102.eu-north-1.compute.internal jenkins[2616]: 2025-10-30 12:59:08.227+0000 [id=68] WARNING h.n.DiskSpaceMonitorD
[ec2-user@ip-172-31-7-102 ~]$
```

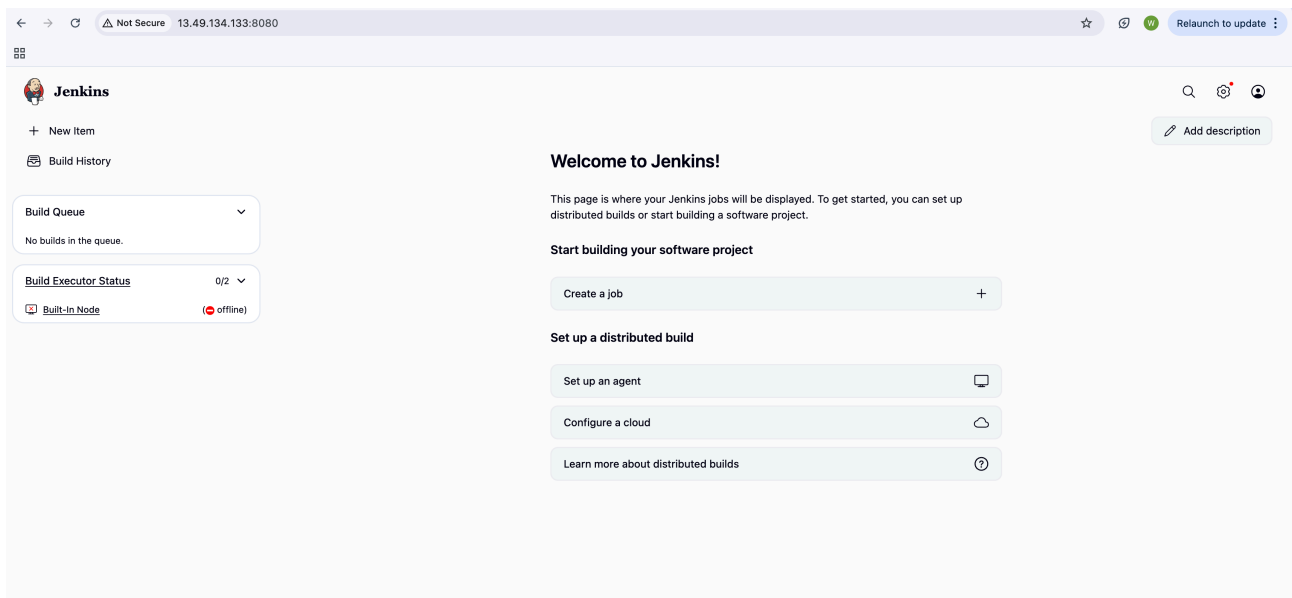


Cat password of Jenkins and copied and past to Jenkins page



Cat command for viewing password

```
cat: /var/lib/jenkins/secrets/initialadminpassword: No such file or directory
[ec2-user@ip-172-31-7-102 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
3c3a5c725dcd4430b80d87060e191af5
[ec2-user@ip-172-31-7-102 ~]$
```



Not Secure 16.16.94.228:8082

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.111

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Server Status](#)
[Manager App](#)
[Host Manager](#)

Developer Quick Start

- [Tomcat Setup](#)
- [Realms & AAA](#)
- [Examples](#)
- [Servlet Specifications](#)
- [First Web Application](#)
- [JDBC DataSources](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

`$CATALINA_HOME/conf/tomcat-users.xml`

In Tomcat 9.0 access to the manager application is split between different users.

[Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)
[Tomcat 9.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/RUNNING.txt`

Developers may be interested in:

[Tomcat 9.0 Bug Database](#)
[Tomcat 9.0 JavaDocs](#)
[Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

Other Downloads

- [Tomcat Connectors](#)
- [Tomcat Native](#)
- [Taglibs](#)
- [Deployer](#)

Other Documentation

- [Tomcat Connectors](#)
- [mod_jk Documentation](#)
- [Tomcat Native](#)
- [Deployer](#)

Get Involved

- [Overview](#)
- [Source Repositories](#)
- [Mailing Lists](#)
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- [Heritage](#)
- [Apache Home](#)
- [Resources](#)

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Ps aux | grep tomcat for checking

```

3C3d5C725aCd4430b80a87060e191dF5
[ec2-user@ip-172-31-7-102 ~]$ ps aux | grep tomcat
root      1465   0.4  7.1 3883928 140160 ?        Sl   12:43   0:06 /opt/java-17/bin/java -Djava.util.logging.config
file:/dev/./urandom -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources
th /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar -Dcatalina.base=/opt
ec2-user  3095   0.0  0.1 222328  2128 pts/0    S+   13:08   0:00 grep --color=auto tomcat
[ec2-user@ip-172-31-7-102 ~]$ sudo netstat -tulnp | grep 8080
tcp6      0      0 :::8080              :::*                  LISTEN      2616/java

```

Yum install docker

```
-bash: docker: command not found
[ec2-user@ip-172-31-7-102 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:25:35 ago on Thu Oct 30 12:58:07 2025.
Dependencies resolved.

Package                               Architecture      Version
-----
Installing:
docker                               x86_64            25.0.13-1.amzn2023.0.1
Installing dependencies:
container-selinux                    noarch            4:2.242.0-1.amzn2023
containerd                           x86_64            2.1.4-1.amzn2023.0.1
iptables-libs                        x86_64            1.8.8-3.amzn2023.0.2
iptables-nft                         x86_64            1.8.8-3.amzn2023.0.2
libcgroup                           x86_64            3.0-1.amzn2023.0.1
libnetfilter_conntrack               x86_64            1.0.8-2.amzn2023.0.2
libnftnl                             x86_64            1.0.1-19.amzn2023.0.2
libnftnl                             x86_64            1.2.2-2.amzn2023.0.2
pigz                                 x86_64            2.5-1.amzn2023.0.3
runc                                  x86_64            1.3.1-1.amzn2023.0.1

Transaction Summary
-----
Install 11 Packages

Total download size: 74 M
Installed size: 280 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm
(9/11): containerd-2.1.4-1.amzn2023.0.1.x86_64.rpm
(10/11): runc-1.3.1-1.amzn2023.0.1.x86_64.rpm
(11/11): docker-25.0.13-1.amzn2023.0.1.x86_64.rpm
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : runc-1.3.1-1.amzn2023.0.1.x86_64
  Installing     : containerd-2.1.4-1.amzn2023.0.1.x86_64
  Running scriptlet: containerd-2.1.4-1.amzn2023.0.1.x86_64
  Installing     : pigz-2.5-1.amzn2023.0.3.x86_64
  Installing     : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  Installing     : libnftnl-1.0.1-19.amzn2023.0.2.x86_64
  Installing     : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
```

Docker started

```
Installed:
  container-selinux-4:2.242.0-1.amzn2023.noarch      containerd-2.1.4-1.amzn2023.0.1.x86_64      docker-25.0.13-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64  libnftnl-1.0.1-19.amzn2023.0.2.x86_64      libnftnl-1.2.2-2.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-10-30 13:24:32 UTC; 1min 17s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 5643 (dockerd)
     Tasks: 9
    Memory: 28.7M
      CPU: 348ms
    CGroup: /system.slice/docker.service
            └─5643 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofil=1048576

Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine:
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.122147493Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.191855280Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.696565239Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.722868267Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.723032786Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.777975586Z"
```

Hello from docker

```
[ec2-user@ip-172-31-7-102 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:56433a6be3fda188089fb548eae3d91df3ed0d6589f7c2656121b911198df065
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
[ec2-user@ip-172-31-7-102 ~]$ █
```

